Boston Scientific

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Boston Scientific Systems

Product Information for Patients

92710144-01

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How to Use this Manual

This *Product Information for Patients (PIP)* provides information about Boston Scientific Systems. Read this manual carefully. If you have any questions or concerns about your System, contact your healthcare provider. In this manual, the term "Boston Scientific Systems" refers to the following Systems:

- Spectra WaveWriter™ System
- Precision Spectra™ System
- Precision Novi™ System
- Precision™ Montage™ MRI System

Guarantees

Boston Scientific Corporation reserves the right to modify, without prior notice, information relating to its products in order to improve their reliability or operating capacity.

Drawings are for illustration purposes only.

Trademark

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Additional Information

For additional information about your system, see the *Information for Patients* manual provided with your System. For instructions on how to use the devices provided with your Boston Scientific Systems, such as your Remote Control or Charger, refer to your *Remote Control Handbook* and *Charger Handbook*. Please note that you will only have a *Charger Handbook* if you have a rechargeable Boston Scientific System.

Contacting Boston Scientific

See the Customer Service section of this manual.

Product Model Numbers

The implantable components of your Boston Scientific System may include the following:

Device Type	Description	Model Number
Stimulators	Precision Spectra™ Implantable Pulse Generator Kit, 32 Contact	SC-1132
	Precision Novi™ Implantable Pulse Generator Kit	SC-1140
	Spectra Wavewriter™ Implantable Pulse Generator Kit	SC-1160
	Precision™ Montage™ MRI Implantable Pulse Generator Kit, 16 Contact	SC-1200

Leads	Linear™ ST Lead Kit, 8 Contact, 30 cm, 50 cm, or 70 cm	SC-2218-xx
	Linear™ ST Trial Lead Kit, 8 Contact, 50 cm or 70 cm	SC-2218-xxE
	Infinion™ 16 Lead Kit, 16 Contact, 50 cm or 70 cm	SC-2316-xx
	Infinion™ 16 Trial Lead Kit, 16 Contact, 50 cm or 70 cm	SC-2316-xxE
	Infinion™ CX Lead Kit, 16 Contact, 50 cm or 70 cm	SC-2317-xx
	Linear™ 3-4 Lead Kit, 8 Contact, 50 cm or 70 cm	SC-2352-xx
	Linear™ 3-4 Trial Lead Kit, 8 Contact, 50 cm	SC-2352-50E
	Linear™ 3-6 Lead Kit, 8 Contact, 30 cm, 50 cm, or 70 cm	SC-2366-xx
	Linear™ 3-6 Trial Lead Kit, 8 Contact, 50 cm	SC-2366-50E
	Avista™ MRI Lead Kit, 8 Contact, 56 cm or 74 cm	SC-2408-xx
	Artisan™ Surgical Lead Kit, 2x8 Contact, 50 cm or 70 cm	SC-8216-xx
	Artisan™ MRI Surgical Lead Kit, 2x8 Contact, 50 cm	SC-8416-xx
	CoverEdge™ 32 Surgical Lead Kit, 4x8 Contact, 50 cm or 70 cm	SC-8336-xx
	CoverEdge™ MRI Surgical Lead Kit, 4x8 Contact, 50 cm or 70 cm	SC-8436-xx
	CoverEdge™ X 32 Surgical Lead Kit, 4x8 Contact, 50 cm or 70 cm	SC-8352-xx
	CoverEdge™ X MRI Surgical Lead Kit, 4x8 Contact, 50 cm or 70 cm	SC-8452-xx
Lead Extensions	Lead Extension Kit, 8 Contact, 25 cm, 35 cm, or 55 cm	SC-3138-xx
Anchors	Clik™ X MRI Anchor	SC-4319
	Clik™ X Anchor	SC-4318
	Clik™ Anchor	SC-4316

Suture Sleeves	Precision™ Suture Sleeve, 1 cm	SC-4300
	Precision™ Suture Sleeve, Split, 1 cm	SC-4301
	Precision™ Suture Sleeve, Tapered, 2.3 cm	SC-4305
	Precision™ Suture Sleeve, Split, 2.3 cm	SC-4306
	Precision™ Suture Sleeve, Segmented, 4 cm	SC-4310
	Precision™ Suture Sleeve, Split, 4 cm	SC-4311
Splitter	D4 Splitter Kit, 2x4 Contact, 25 cm	SC-3304-25
	W4 Splitter Kit, 2x4 Contact, 25 cm	SC-3354-25
	Splitter Kit, 2x8 Contact, 30 cm	SC-3400-30
Adapters/Connectors	Precision™ Adapter S8, 8 Contact, 15 cm or 55 cm	SC-9208-xx
	Precision™ Adapter M8, 8 Contact, 15 cm or 55 cm	SC-9218-xx
	Precision™ Connector M-1, 8 Contact, 35 cm, 55 cm, or 70 cm	SC-9004-xx
Medical Adhesive	Medical Adhesive	SC-4320
Port Plug	Precision Spectra™ IPG Port Plug	SC-4401
XX refers to length		

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Description

The Boston Scientific System uses electrical pulses to stimulate the spine and peripheral nerves. The System includes parts that are implanted in the body. These parts include the Leads, Lead Extensions, and the Stimulator. In addition, the System includes external devices that are used to manage the implanted Stimulator. These external devices include a Remote Control and a Charging System. A Remote Control is used to adjust stimulation. A Charging System is used to charge the battery of the Stimulator. More information about these devices, can be found in the Patient Manuals provided with the devices.

Intended Use

The intended use of the System is to treat chronic pain by stimulating the spinal cord or peripheral nerves.

Patient Population

Patients who have chronic, intractable pain.

Monitoring the System

A Remote Control device is provided with the System. The Remote Control is a device that allows patients to communicate with their Stimulators through settings, which are called Programs. Programs may also be grouped together to create Schedules. The Remote Control allows the patient to manage the programming that the healthcare provider created to deliver therapy. The patient can use the Remote Control to turn stimulation on or off. The patient can also view information about the implanted Stimulator, such as the following:

- The battery level of the Stimulator: if the patient has a rechargeable Stimulator, the Remote Control will display the level of the Stimulator battery. The patient uses this information to decide when to recharge the Stimulator.
- Important messages about the Stimulator, such as whether stimulation is on or off.
- Important settings for the Stimulator, such as settings that should be used before the patient undergoes an MRI scan.

Only use the Remote Control provided with the Boston Scientific System. Other similar models of the Remote Control will not function with the Boston Scientific System. Refer to the *Remote Control Handbook* for more details.

Maintaining the System

The rechargeable system must be charged. The patient must charge the Stimulator battery periodically to continue receiving stimulation. When the Stimulator battery is running low, the Remote Control will display messages to charge the Stimulator. A Charging system is provided with the rechargeable Systems. Refer to the *Charging Handbook* for more details.

The non-rechargeable Systems do not require charging. The battery in these Stimulators will continue to provide stimulation until the battery is used up. When the battery is used up, surgery is needed to replace the

Stimulator. The Remote Control will display messages when the Stimulator battery is low. Refer to the *Remote Control Handbook* for more details.

Post-Operative Information

During the two weeks following surgery, it is important to use extreme care so that healing can occur. Healing will secure the implanted components and close the surgical incisions.

- Do not lift more than 2.5 kilograms (5 pounds).
- Do not engage in rigorous physical activity such as twisting, bending, or climbing.
- If new Leads were implanted, do not raise your arms above your head.

Temporarily, there may be some pain in the area of the implant as the incisions heal. If discomfort continues beyond two weeks, contact your physician.

If you notice excessive redness around the wound areas during this time, contact your physician to check for infection and administer proper treatment. In rare cases, adverse tissue reaction to implanted materials can occur during this period.

Be sure to consult your physician before making any lifestyle changes due to decreases in pain.

Adverse Events

Potential risks are involved with any surgery. Report any serious incident that occurs in relation to this device to Boston Scientific and to the relevant local regulatory authority for medical devices in your country. For customers in Australia, report any serious incident that occurs in relation to this device to Boston Scientific and to the Therapeutic Goods Administration (https://www.tga.gov.au).

The possible risks include the following:

- The Lead(s) which deliver stimulation may move from their original implanted location, resulting in undesirable changes in stimulation and subsequent reduction in pain relief.
- System failure, which can occur at any time due to random failure(s) of the components or the battery. These events, which may include battery leakage, device failure, Lead breakage, hardware malfunctions, loose connections, electrical shorts or open circuits and Lead insulation breaches, can result in ineffective pain control.
- Your body may react negatively to the materials used to manufacture the Stimulator or the Leads. You may notice redness, warmth or swelling of the implant area. Tissue reaction to implanted materials can occur. In some cases, the formation of reactive tissue around the Lead in the epidural space can result in delayed onset of spinal cord compression and neurological/sensory deficit, including paralysis. Time to onset is variable, possibly ranging from weeks to years after implant.
- The skin over your implant may become thin and increasingly tender over time. A seroma may be formed.
- The most common surgical procedural risks are temporary pain at the implant site and infection. However,

since the Leads are placed in the epidural space, there is a small risk that spinal fluid may leak from the Lead insertion site following surgery. Very rarely, you may develop an internal blood clot (hematoma) or blister (seroma); or you may experience epidural hemorrhage or paralysis. Your spinal cord may become compressed.

- External sources of electromagnetic interference may cause the device to malfunction and affect stimulation.
- Exposure to magnetic resonance imaging (MRI) can result in discomfort or injury due to heat near the Stimulator or the leads, tugging or vibration of the implanted system, induced stimulation, damage to the device requiring its replacement and may distort the image needed for diagnosis.
- Undesirable stimulation may occur over time due to cellular changes in tissue around the electrodes, changes in electrode position, loose electrical connections and/or Lead failure.
- You may experience painful electrical stimulation of your chest wall as a result of stimulation of certain nerve roots several weeks after surgery.
- Over time, your implant may move from its original position.
- You may experience weakness, clumsiness, numbness or pain below the level of implantation.
- You may experience persistent pain at the Stimulator or Lead site.

In the event of painful stimulation, uncomfortable stimulation, or suspected device malfunction, use the Remote Control to turn stimulation off, or reduce stimulation level, and contact your healthcare provider. In any event, contact your physician and inform him/her.

General Warnings

Refer to the *Information for Patients IFU* for additional information on the warnings provided below.

- Unauthorized changes to the medical devices are not allowed.
- If you are using a rechargeable Stimulator:
  - The Charger may become warm while charging your Stimulator. Use the Charger with the Charging Belt, or an Adhesive Patch to avoid a burn.
  - Do not charge your device while sleeping. This may result in a burn.
  - Stop charging and contact your healthcare provider if you experience pain or discomfort.
- For Spinal Cord Stimulator (SCS) Systems, you may be able to have an MRI examination when specific conditions are met. These conditions are defined in the supplemental physician manual *"ImageReady™ MRI Head Only Guidelines for Precision Spectra™, ImageReady™ MRI Head Only Guidelines for Spectra WaveWriter™, and ImageReady™ MRI Full Body Guidelines for Precision™ Montage™ MRI Spinal Cord Stimulator Systems"*. These manuals can be found on the Boston Scientific website (www.bostonscientific-elabeling.com).

- MRI-conditional labeling does not apply to Peripheral Nerve Stimulation. For Peripheral Nerve Stimulation (PNS) Systems, do not be exposed to an MRI.
- Boston Scientific external components (such as, the Trial Stimulator, Remote Control, Battery Charger) are **MR Unsafe**. Do not take them into any MRI environment.
- The safety and effectiveness of stimulation has not been established for pediatric use.
- As a patient, you should not have any form of diathermy either as treatment for a medical condition or as part of a surgical procedure. The high energy and heat generated by diathermy can be transferred through your Stimulator system, causing tissue damage at the Lead site and, possibly, severe injury or death. The Stimulator, whether it is turned on or off, may be damaged.
- Using other implantable stimulation devices at the same time as the Boston Scientific Stimulator may interfere with the operation of the devices. Examples of other implantable stimulation devices include pacemakers or cardioverter defibrillators. If you must use other implantable stimulation devices at the same time as the Stimulator, careful programming of each system is necessary.
- Burns may result if your Stimulator is ruptured or pierced and patient tissue is exposed to battery chemicals.
- Changes in posture or sudden movements may decrease, or increase the perceived stimulation level. Keep the Remote Control with you at all times and turn the stimulation down or off before making posture changes. If unpleasant sensations occur, the stimulation should be turned off immediately. If using therapy that does not produce a sensation, postural changes are less likely to affect you.

- Strong electromagnetic fields can cause temporary changes in your stimulation. Be careful around equipment with strong electromagnetic fields.

Precautions

- Physician training is required.
- Turn off stimulation before operating an automobile, other motorized vehicle, or any potentially dangerous machinery/equipment. Sudden stimulation changes, if they occur, may distract you from attentive operation of the vehicle or equipment. If using therapy that does not produce a sensation, sudden stimulation changes are less likely to affect you.
- Avoid touching the incisions or Stimulator site. Never attempt to change the orientation of the Stimulator or turn over the Stimulator. If the Stimulator flips over in your body, it may be unable to communicate with the Remote Control or Clinician Programmer. If the rechargeable Stimulator flips over in your body, then it cannot be charged. If stimulation cannot be turned on after charging, contact your healthcare provider. If you notice a change in the appearance of the skin at the Stimulator location, such as the skin becoming thin over time, contact your healthcare provider.
- In some instances a Lead can move from its original location, and stimulation at the intended pain site can be lost. If this occurs, consult your physician.
- Implants can fail at any time. If the device stops working even after complete charging (up to four hours), turn off the Stimulator and contact your physician.

- The Stimulator should be removed from the body before cremation and returned to Boston Scientific.
- Interference caused by cell phones is not anticipated, but the full effects of interaction with cell phones are unknown at this time. Portable RF communications equipment (for example, mobile phones) should be kept a minimum distance of 6 inches (15 cm) from the area of the implanted device. If interference does occur, move the cell phone away from the Stimulator or turn off the phone. If there is a concern or a problem is encountered, contact your healthcare provider.

Medical Devices and Therapies

The following medical therapies or procedures may turn stimulation off or may cause permanent damage to the Stimulator, particularly if used in close proximity to the device:

- lithotripsy - high-output sound or shock waves often used to treat gall stones and kidney stones
- electrocautery - the use of a heated electric probe to stop bleeding during surgery
- external defibrillation - the use of electrically charged paddles to restart the heart in an emergency
- radiation therapy - ionizing energy commonly used to treat cancer. Any damage to the device by radiation may not be immediately detectable.
- ultrasonic scanning - very high frequency sound waves used to produce images of internal organs or tissue for diagnostic purposes

- high-output ultrasound - high frequency sound waves which may be applied as physical therapy to treat certain bone/muscle injuries, or for muscle stimulation, or to improve blood flow

X-ray and CT scans may damage the Stimulator if stimulation is on. X-ray and CT Scans are unlikely to damage the Stimulator if stimulation is turned off.

Before having procedures, medical therapies, or diagnostics, have your healthcare provider call our Customer Service department for proper instructions. Refer to “Contacting Boston Scientific” in this manual for contact information for your locality.

Materials

The table below shows the materials and substances that contact the body during the implant procedure:

Device Type	Implantable Material	Percent Weight
Stimulators	Titanium	18 % to 57 %
	Epoxy	42 % to 82 %
	Silicone	Up to 3 %
Stimulator Template	Stainless Steel	100 %
Stimulator Port Plug	Thermoplastic	100 %

Leads	Thermoplastic	65 % to 94 %
	Platinum - Iridium	6 % to 26 %
	Silicone	0 % to 19 %
	Epoxy	0.05 % to 0.36 % (< 1 %)
Suture Sleeve	Silicone	100 %
MR Conditional Suture Sleeve	Silicone doped with barium sulfate	100 %
Lead Extension/Splitter	Silicone	60 % to 66 %
	Thermoplastic	34 % to 40 %
	Epoxy	< 1
Adapters/Connectors	Silicone	37 % to 74 %
	Thermoplastic	26 % to 63 %
	Epoxy	< 1
Clik Anchors	Silicone	95 %
	Thermoplastic	5 %
Clik X Anchor	Silicone	100 %
Clik X MRI Anchor	Silicone doped with barium sulfate	100 %
Tunneling Tool Shaft	Stainless Steel	100 %
Tunneling Tool Straw	PTFE	100 %
Medical Adhesive	Medical Adhesive Silicone	100 %

The IPG Template, Insertion Needle, Entrada Needle, Lead Blank, and Tunneling Tool may contain cobalt: CAS No. 7440-48-4; EC No. 231-158-0. Defined as a 1B carcinogen and reproductive toxicant according to the European Commission in a concentration above 0.1 % weight by weight.

Note: *The IPG Template, Insertion Needle, Entrada Needle, Lead Blank, and Tunneling Tool are made with stainless steel which may contain cobalt. Current scientific evidence supports that metal alloys containing cobalt used in medical devices do not cause an increased risk of cancer or adverse reproductive effects.*

Intended Performance

Trial Stimulator

The stimulation pulse shall meet the requirements for charge balance and amplitude while stimulation is on.

Other External Devices

Failure of the external electrical components will not result in an unacceptable risk to the user.

Expected Lifetime of Implant

See the *Information for Patients IFU* for information on the lifetime of the devices.

Customer Service

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Note: *Phone numbers and fax numbers may change. For the most current contact information, please refer to our website at http://www.bostonscientific-international.com/ or write to the following address:*

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Valencia, CA 91355 USA

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Boston Scientific
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Argentina Local Contact

Para obtener información de contacto de Boston Scientific Argentina SA, por favor, acceda al link www.bostonscientific.com/arg

Brazil Local Contact

Para informações de contato da Boston Scientific do Brasil Ltda, por favor, acesse o link www.bostonscientific.com/bra

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